# Revisiting Graph Attention Networks: A Reproduction Study with phi-Attention Scoring

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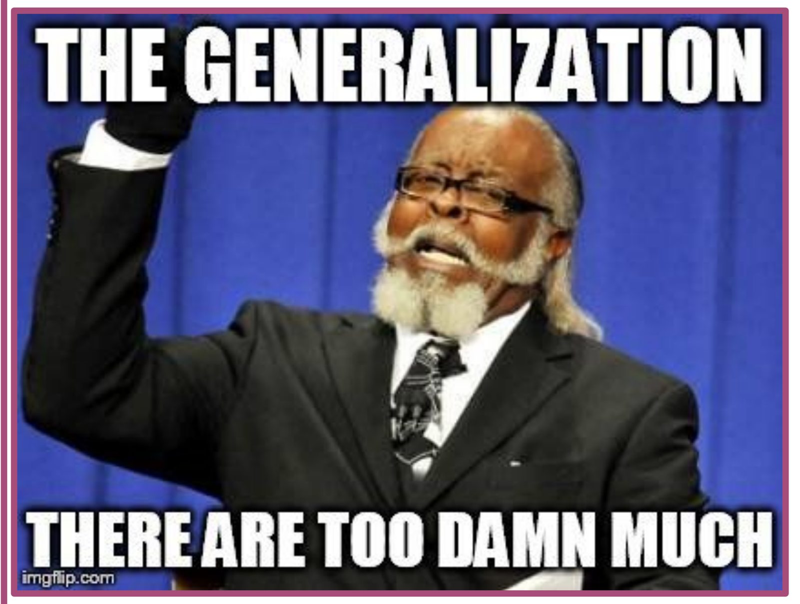
**GAT Paper QR Code**

## Motivation

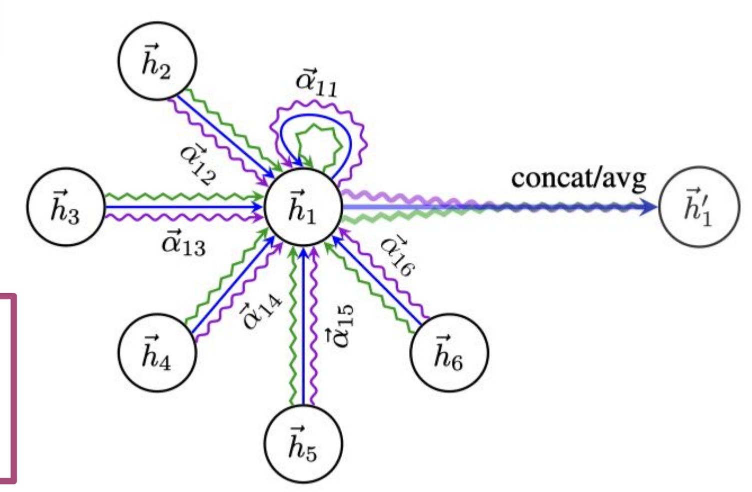
**The GATv1 Paper…**





$$e_{ij} = a(\mathbf{W}\vec{h}_i, \mathbf{W}\vec{h}_j)$$

$$\alpha_{ij} = \mathrm{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik})}.$$




## Clarification and Methodology

**Cora** is a citation network where nodes represent scientific papers and edges represent citation relationships.
**Accuracy** measures the proportion of correctly classified nodes on the test set.

**Comparison**

$$e_{ij} = \vec{a}^T \left(\mathrm{LeakyReLU}\left(W\left[\vec{h}_i \| \vec{h}_j\right]\right)\right)$$

$$e_{ij} = \vec{a}^T \left(\mathrm{LeakyReLU}\left(W\left[\vec{h}_i \| \vec{h}_j \| \phi_{ij}\right]\right)\right)$$

**GAT V2**

Whether incorporate structural information improve classification accuracy

$$e_{ij} = \mathrm{LeakyReLU}\left(\vec{a}^T\left[W\vec{h}_i \,\|\, W\vec{h}_j \,\|\, \phi_{ij}\right]\right)$$

$$\phi_{ij} = \left[\frac{\deg(j)}{\deg_{\max}},\ \frac{|\mathcal{N}_i \cap \mathcal{N}_j|}{|\mathcal{N}_i \cup \mathcal{N}_j|},\ \sum_{k \in \mathcal{N}_i \cup \mathcal{N}_j} \frac{1}{\log(\deg(k))}\right]$$

**Alternative** $\Phi_{ij}$

$$e_{ij} = \mathrm{LeakyReLU}\left(\vec{a}^T\left[W\vec{h}_i \,\|\, W\vec{h}_j\right]\right)$$

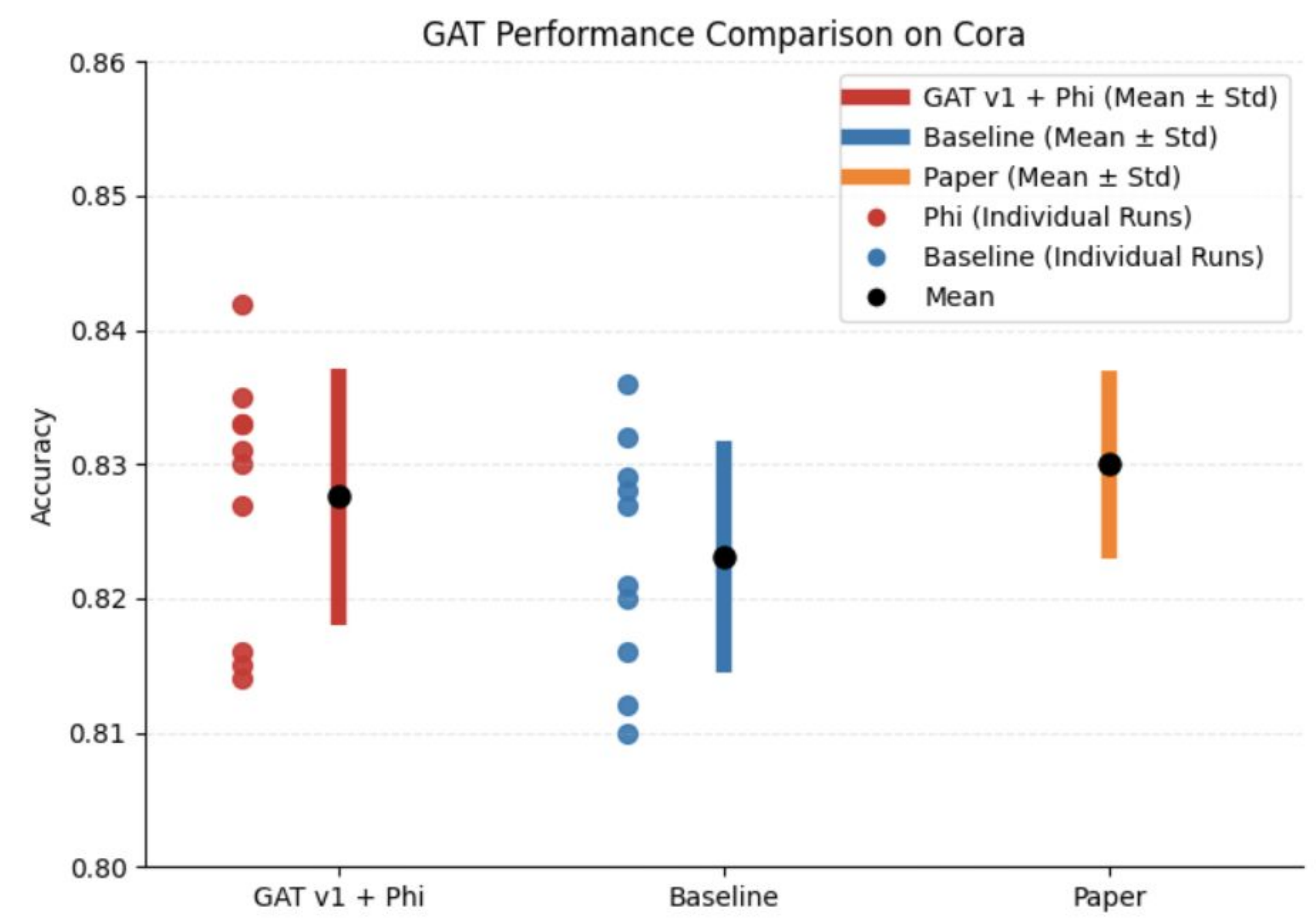
**Baseline**

## Reproduction Results And Modification

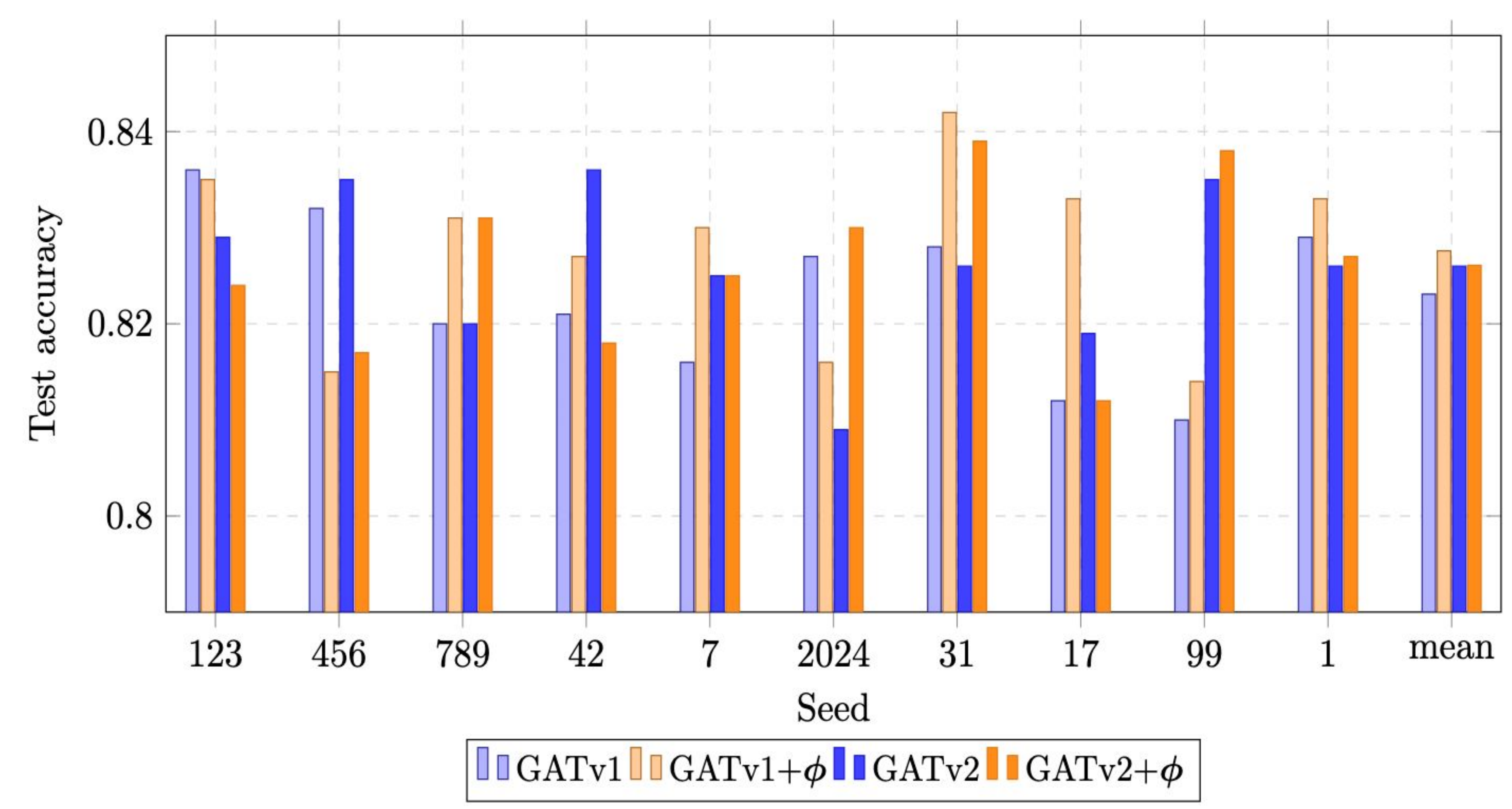
### Baseline - GAT v1

The original GAT implementation only provides official support for the Cora Dataset, therefore reproduce using the authors code and data as the baseline.

- **Implementation Adaptation** The code is based on TensorFlow 1.x, which is incompatible with modern environments. We fixed it.
- **Experimental Results** We perform multiple runs to account for stochasticity and obtain **the baseline result.** This result falls within the range reported in the original paper, confirming successful reproduction.




**GAT v1 with ϕ outperforms**




## Conclusion & Discussion

**Overall Improvement**: Our updated version of GAT outperforms the original paper for the Cora Dataset, indicating that the design of ϕ is useful and successful.
**Computational Efficiency**: Introducing ϕ leads to O(E*d) of memory and computing complexity, which is small compared to GAT attention computation. So our design doesn't severely influence the efficiency of the original algorithm.
**Why ϕ Helps**:The performance gain from ϕ suggests that structural priors such as neighborhood similarity and node centrality provide complementary information beyond node features. In homophilous graphs like Cora, nodes with similar labels tend to share neighbors, making structural similarity a strong signal for attention weighting.

$$\frac{\deg(j)}{\deg_{\max}}$$

means: **how highly connected paper $j$ is in the citation network.**

$$\frac{|\mathcal{N}_i \cap \mathcal{N}_j|}{|\mathcal{N}_i \cup \mathcal{N}_j|}$$

means: **how many citation-neighbors papers $i$ and $j$ share.**

$$\sum_{k \in \mathcal{N}_i \cup \mathcal{N}_j} \frac{1}{\log(\deg(k))}$$

means: **how much specific local citation context surrounds papers $i$ and $j$, giving more weight to less-connected papers.**

## Future Work


**? Citeseer & Pubmed**

**Post-2018 advancements**

**Rethinking Graph Transformers with Spectral Attention**

**Edge-Featured Graph Attention Network**

**Structure-Aware Transformer for Graph Representation Learning**

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## Reference

Petar Velickovi ˇ c, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In International Conference on Learning Representations, 2018.

## Acknowleage

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